

Preprocessing

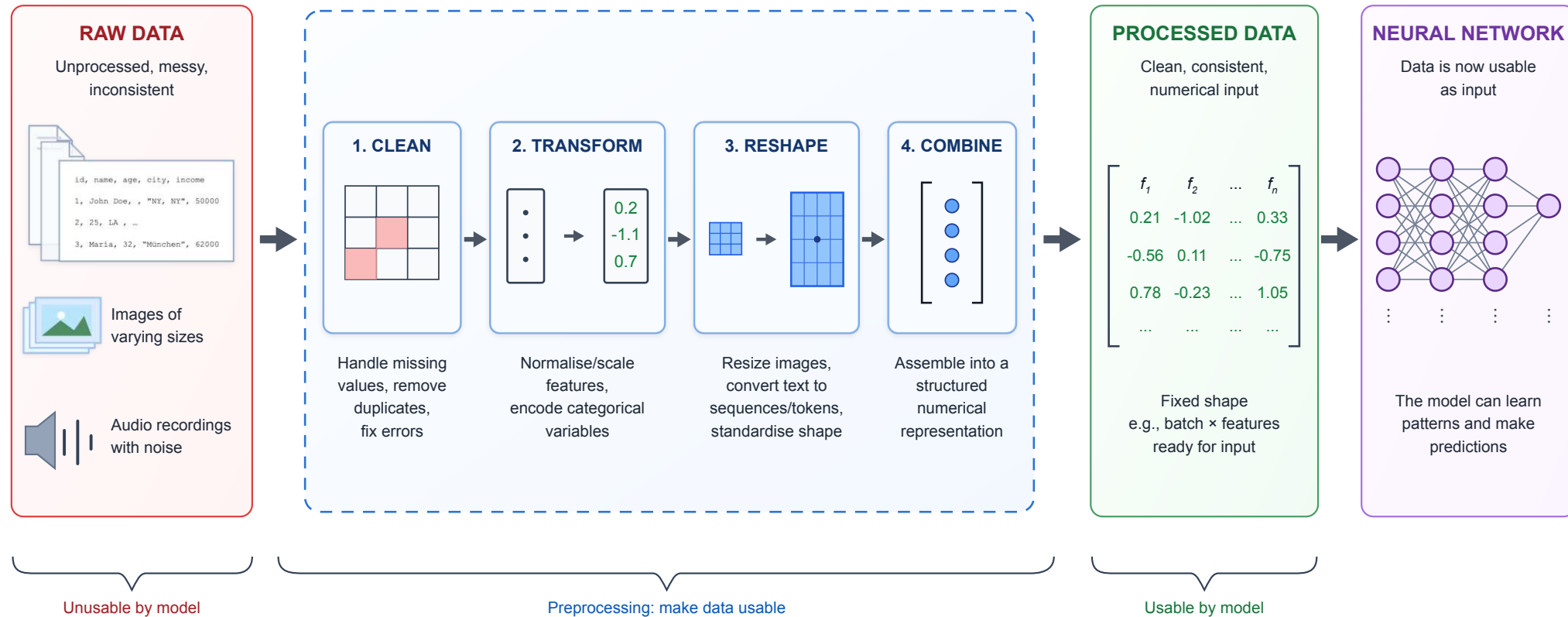
ACTL3143 & ACTL5111 Deep Learning for Actuaries
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Deep learning project life cycle

1. Define objectives
2. *Collect & label data*
3. Clean & preprocess data (opt. *feature engineering*)
4. Design network architecture
5. Train and validate model
6. Hyperparameter tuning
7. Test and evaluate performance
8. *Deploy to production*
9. *Monitor and iterate*

(We won't cover the italicised items in this course, though they are very important.)

Overview



Preprocessing refers to the transformations of the raw data before input to the network. Some steps are specific to neural networks (e.g. standardising continuous variables), others are necessary for every model (e.g. encoding categorical variables, handling missing data).

Imports needed for these demos

```
1 import random
2 from pathlib import Path
3
4 import joblib
5 import matplotlib.pyplot as plt
6 import numpy as np
7 import pandas as pd
8
9 from keras.models import Sequential
10 from keras.layers import Dense, Input
11 from keras.callbacks import EarlyStopping
12
13 import sklearn
14 from sklearn import set_config
15 from sklearn.model_selection import train_test_split
16 from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, StandardScaler
17 from sklearn.impute import SimpleImputer
18 from sklearn.compose import make_column_transformer
19 from sklearn.pipeline import make_pipeline
20 from sklearn.linear_model import LinearRegression
```

Lecture Outline

- **Standardising the numerical covariates**
- Handling categorical variables
- Missing values and rare observations
- Procedures to keep your data organised

California housing dataset

```
1 features, target = sklearn.datasets.fetch_california_housing(
2     as_frame=True, return_X_y=True)
3 features
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86
...	...	...	...	...	...	...	...
20638	1.8672	18.0	5.329513	1.171920	741.0	2.123209	39.41
20639	2.3886	16.0	5.254717	1.162264	1387.0	2.616981	39.37

20640 rows × 8 columns

Pandas will assign or guess dtypes

Can look at the data types of the features.

```
1 features.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   MedInc          20640 non-null  float64
1   HouseAge        20640 non-null  float64
2   AveRooms        20640 non-null  float64
3   AveBedrms       20640 non-null  float64
4   Population      20640 non-null  float64
5   AveOccup        20640 non-null  float64
6   Latitude        20640 non-null  float64
7   Longitude       20640 non-null  float64
dtypes: float64(8)
memory usage: 1.3 MB
```

Here, all the features were continuous variables

Pandas can guess wrong though

For example, Australian postcodes are four digits and some start with `0` (e.g. Darwin is `0800`). A CSV reader just sees digits, guesses a number, and silently drops the leading zero — giving the column the wrong *type* **and** the wrong *value*.

Default inference:

```
1 df = pd.read_csv("data/postcodes.csv")
2 df
```

	suburb	postcode
0	Darwin	800
1	Sydney	2000
2	Perth	6000

```
1 df.dtypes
```

```
suburb      str
postcode    int64
dtype: object
```

The raw file keeps the leading zero:

```
1 print(Path("data/postcodes.csv").read_text())
```

```
suburb,postcode
Darwin,0800
Sydney,2000
Perth,6000
```

Typed at read time:

```
1 df = pd.read_csv("data/postcodes.csv",
2                 dtype={"postcode": "string"})
3 df
```

	suburb	postcode
0	Darwin	0800
1	Sydney	2000
2	Perth	6000

```
1 df.dtypes
```

```
suburb      str
postcode    string
dtype: object
```

Splitting into train, validation and test sets

```
1 X_main, X_test_raw, y_main, y_test = train_test_split(  
2     features, target, test_size=0.2, random_state=1  
3 )  
4 X_train_raw, X_val_raw, y_train, y_val = train_test_split(  
5     X_main, y_main, test_size=0.25, random_state=1  
6 )
```

Note that we really cannot allow there to be duplicate rows in the training, validation, or test sets. If there are, we will have data leakage between the sets, which will lead to overfitting and poor generalisation.

Checking for leaks

We can look for rows that appear in multiple of the datasets.

```
1 train_val_overlap = pd.merge(X_train_raw, X_val_raw, how='inner')
2 train_test_overlap = pd.merge(X_train_raw, X_test_raw, how='inner')
3 val_test_overlap = pd.merge(X_val_raw, X_test_raw, how='inner')
4
5 print(f"Number of duplicated rows between train and val: {len(train_val_overlap)}")
6 print(f"Number of duplicated rows between train and test: {len(train_test_overlap)}")
7 print(f"Number of duplicated rows between val and test: {len(val_test_overlap)}")
```

Number of duplicated rows between train and val: 0

Number of duplicated rows between train and test: 0

Number of duplicated rows between val and test: 0

An earlier check is to check that there are no duplicate rows in the dataset.

```
1 features.duplicated().sum()
```

np.int64(0)

We rescaled every column

```

1 scaler = StandardScaler()
2 scaler.fit(X_train_raw)
3
4 X_train = scaler.transform(X_train_raw)
5 X_val = scaler.transform(X_val_raw)
6 X_test = scaler.transform(X_test_raw)
7
8 X_train

```

```

array([[ 0.15, -0.76,  0.26, ..., -0.01, -0.47,  0.69],
       [-1.79, -1.48,  0.48, ..., -0.08, -0.44,  1.33],
       [-0.97, -0.13, -0.67, ...,  0.05, -0.86,  0.65],
       ...,
       [ 0.11, -0.84, -0.54, ..., -0.03, -0.81,  0.6 ],
       [-0.82,  0.99, -0.03, ..., -0.03,  0.54, -0.11],
       [ 0.9 , -1.56,  0.66, ...,  0.07, -0.75,  0.97]], shape=(12384, 8))

```

That is because the neural network expects continuous inputs to be normalised to help with the training procedure.

Note

Notice that the output of `X_train` here looks a bit different to `X_train_raw` earlier? We'll touch on this in a couple of slides.

Scikit-learn preprocessing methods

- `fit`: learn the parameters of the transformation
- `transform`: apply the transformation
- `fit_transform`: learn the parameters and apply the transformation

<code>fit</code>	<code>fit_transform</code>
------------------	----------------------------

```

1 scaler = StandardScaler()
2 scaler.fit(X_train_raw)
3 X_train = scaler.transform(X_train_raw)
4 X_val = scaler.transform(X_val_raw)
5 X_test = scaler.transform(X_test_raw)
6
7 print(X_train.mean(axis=0))
8 print(X_train.std(axis=0))
9 print(X_val.mean(axis=0))
10 print(X_val.std(axis=0))

```

```

[ 1.14e-16 -1.08e-16 -2.82e-16 -4.45e-17 -6.89e-18
-4.02e-17  1.14e-15
 1.75e-15]
[1.  1.  1.  1.  1.  1.  1.  1.]
[ 0.01  0.01  0.   -0.01 -0.    0.01  0.   -0.01]
[1.01  1.    0.81  0.81  0.98  0.84  0.99  1.   ]

```

Dataframes & arrays

1 X_test_raw

	MedInc	HouseAge	AveRooms
4712	3.2500	39.0	4.503205
2151	1.9784	37.0	4.988584
...	...	...	...
6823	4.8750	42.0	5.347985
11878	2.7054	52.0	5.741214

4128 rows × 8 columns

1 X_test

```
array([[ -0.33,  0.83, -0.34, ..., -0.11,
        -0.73,  0.6 ],
       [ -1.   ,  0.67, -0.16, ..., -0.04,
         0.54, -0.1 ],
       [  0.07,  1.38, -0.35, ...,  0.06,
         0.98, -1.42],
       ...,
       [  0.62, -0.21, -0.16, ...,  0.05,
         0.92, -1.4 ],
       [  0.53,  1.07, -0.03, ..., -0.   ,
        -0.72,  0.73],
       [-0.62,  1.86,  0.11, ..., -0.02,
        -0.77,  1.09]], shape=(4128, 8))
```

i Note

By default, when you pass `sklearn` a DataFrame it returns a `numpy` array.

Keep as a DataFrame

From **scikit-learn 1.2**:

```
1 set_config(transform_output="pandas")
2
3 imp = SimpleImputer()
4
5 X_train = imp.fit_transform(X_train_raw)
6 X_val = imp.transform(X_val_raw)
7 X_test = imp.transform(X_test_raw)
```

1 X_test

	MedInc	HouseAge	AveRooms
4712	3.2500	39.0	4.503205
2151	1.9784	37.0	4.988584
...	...	...	...
6823	4.8750	42.0	5.347985
11878	2.7054	52.0	5.741214

4128 rows × 8 columns

From SimpleImputer's docs:

Replace missing values using a descriptive statistic (e.g. mean, median, or most frequent) along each column, or using a constant value... default='mean'

Lecture Outline

- Standardising the numerical covariates
- **Handling categorical variables**
- Missing values and rare observations
- Procedures to keep your data organised

French motor dataset

```

1  # Download the dataset if we don't have it already.
2  if not Path("data/freq_data.csv").exists():
3      freq = sklearn.datasets.fetch_openml(data_id=41214, as_frame=True).frame
4      freq.to_csv("data/freq_data.csv", index=False)
5  else:
6      freq = pd.read_csv("data/freq_data.csv")
7  freq.info()

```

```

<class 'pandas.DataFrame'>
RangeIndex: 678013 entries, 0 to 678012
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  --
0   IDpol            678013 non-null float64
1   ClaimNb          678013 non-null float64
2   Exposure         678013 non-null float64
3   Area             678013 non-null str
4   VehPower         678013 non-null float64
5   VehAge           678013 non-null float64
6   DrivAge          678013 non-null float64
7   BonusMalus       678013 non-null float64
8   VehBrand         678013 non-null str
9   VehGas           678013 non-null str
10  Density          678013 non-null float64

```

The data

1 freq

	IDpol	ClaimNb	Exposure	Area	VehPower	VehAge	DrivAge	Bon
0	1.0	1.0	0.10000	D	5.0	0.0	55.0	50.0
1	3.0	1.0	0.77000	D	5.0	0.0	55.0	50.0
2	5.0	1.0	0.75000	B	6.0	2.0	52.0	50.0
...	...	...	...	...	...	...	...	...
678010	6114328.0	0.0	0.00274	D	6.0	2.0	45.0	50.0
678011	6114329.0	0.0	0.00274	B	4.0	0.0	60.0	50.0
678012	6114330.0	0.0	0.00274	B	7.0	6.0	29.0	54.0

678013 rows × 12 columns

Data dictionary

- **IDpol**: policy number (unique identifier)
- **ClaimNb**: number of claims on the given policy
- **Exposure**: total exposure in yearly units
- **Area**: area code (categorical, ordinal)
- **VehPower**: power of the car (categorical, ordinal)
- **VehAge**: age of the car in years
- **DrivAge**: age of the (most common) driver in years
- **BonusMalus**: bonus-malus level between 50 and 230 (with reference level 100)
- **VehBrand**: car brand (categorical, nominal)
- **VehGas**: diesel or regular fuel car (binary)
- **Density**: of inhabitants per km² in the city of the living place of the driver
- **Region**: regions in France (prior to 2016)

We have $\{(\mathbf{x}_i, y_i)\}_{i=1, \dots, n}$ for $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \mathbb{N}_0$. Assume the distribution

$$Y_i \sim \text{Poisson}(\lambda(\mathbf{x}_i))$$

We have $\mathbb{E}Y_i = \lambda(\mathbf{x}_i)$. The NN takes $\mathbf{x}_i$ & predicts $\mathbb{E}Y_i$.

```
1 freq = freq.drop("IDpol", axis=1)
```

Split the data

```

1 X_train_raw, X_test_raw, y_train, y_test = train_test_split(
2     freq.drop("ClaimNb", axis=1), freq["ClaimNb"], random_state=2023)
3 X_train_raw = X_train_raw.reset_index(drop=True)
4 X_test_raw = X_test_raw.reset_index(drop=True)
5 X_train_raw

```

	Exposure	Area	VehPower	VehAge	DrivAge	BonusMalus	VehBrand
0	0.01	A	4.0	0.0	70.0	50.0	B12
1	1.00	C	6.0	11.0	36.0	50.0	B3
2	0.08	D	7.0	9.0	35.0	50.0	B2
...	...	...	...	...	...	...	...
508506	0.22	E	7.0	21.0	32.0	90.0	B5
508507	1.00	C	7.0	15.0	51.0	50.0	B1
508508	0.42	D	6.0	1.0	37.0	54.0	B2

508509 rows × 10 columns

What values do we see in the data?

```
1 X_train_raw["Area"].value_counts()
2 X_train_raw["VehBrand"].value_counts()
3 X_train_raw["VehGas"].value_counts()
4 X_train_raw["Region"].value_counts()
```

Area

```
C      144156
D      113532
E      102791
A       78017
B       56571
F       13442
Name: count, dtype: int64
```

VehGas

```
Regular    259422
Diesel     249087
Name: count, dtype: int64
```

VehBrand

```
B12      124490
B1       122200
B2       120155
...
B11       10218
B13        9160
B14        2998
Name: count, Length: 11, dtype: int64
```

Region

```
R24      120597
R82       63555
R93       59627
...
R21        2283
R42        1634
R43        1024
Name: count, Length: 22, dtype: int64
```

Nominal categorical variables

These are categorical variables which you cannot order in a 'default' sensible way.

```
1 gas = X_train_raw[["VehGas"]]
```

The original data:

```
1 gas.head()
```

	VehGas
0	Regular
1	Regular
2	Diesel
3	Regular
4	Diesel

One-hot encoding

```
1 enc = OneHotEncoder(sparse_output=False)
2
3 X_train = enc.fit_transform(gas)
4 X_train.head()
```

	VehGas_Diesel	VehGas_Regular
0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	0.0	1.0
4	1.0	0.0

Dummy encoding

```
1 enc = OneHotEncoder(
2     drop="first",
3     sparse_output=False)
4 X_train = enc.fit_transform(gas)
5 X_train.head()
```

	VehGas_Regular
0	1.0
1	1.0
2	0.0
3	1.0
4	0.0

⚠ Warning

Collinearity is not a big deal for neural networks. However if you want to reuse the same `X_train` data for a (generalised) linear regression, then you probably want to dummy-encode here to avoid needing two different preprocessing regimes for the two competing models.

Ordinal categorical variables

These are categorical variables which have a natural order to them.

```
1 enc = OrdinalEncoder()
2 enc.fit(X_train_raw[["Area"]])
3 enc.categories_
```

```
[array(['A', 'B', 'C', 'D', 'E', 'F'], dtype=object)]
```

```
1 for i, area in enumerate(enc.categories_[0]):
2     print(f"The Area value {area} gets turned into {i}.")
```

The Area value A gets turned into 0.

The Area value B gets turned into 1.

The Area value C gets turned into 2.

The Area value D gets turned into 3.

The Area value E gets turned into 4.

The Area value F gets turned into 5.

```
1 print(f"In other words, {' < '.join(enc.categories_[0])}, and ")
2 print(" = ".join(f"{r} - {l}" for l, r in zip(enc.categories_[0][: -1], enc.categories_[0]
```

In other words, $A < B < C < D < E < F$, and

$B - A = C - B = D - C = E - D = F - E$

Ordinal encoded values

```
1 X_train = enc.transform(X_train_raw[["Area"]])
2 X_test = enc.transform(X_test_raw[["Area"]])
```

```
1 X_train_raw[["Area"]].head()
```

	Area
0	A
1	C
2	D
3	E
4	B

```
1 X_train.head()
```

	Area
0	0.0
1	2.0
2	3.0
3	4.0
4	1.0

We could train on this `X_train`, or on the previous `X_train`, but each only contains one feature from the dataset. How do we add the continuous variables back in? Use a sklearn *column transformer* for that.

Transform all columns in one hit I

```

1 ct = make_column_transformer(
2     (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
3     (OrdinalEncoder(), ["Area"]),
4     remainder=StandardScaler()
5 )
6
7 X_train = ct.fit_transform(X_train_raw)

```

1 X_train_raw

	Exposure	Area	VehPower	VehBrand
0	0.01	A	4.0	0.0
...	...	...	...	...
508508	0.42	D	6.0	1.0

508509 rows × 10 columns

1 X_train

	onehotencoder__VehGas_Regul
0	1.0
...	...
508508	0.0

508509 rows × 39 columns

Transform all columns in one hit II

```

1 ct = make_column_transformer(
2     (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
3     (OrdinalEncoder(), ["Area"]),
4     remainder=StandardScaler(),
5     verbose_feature_names_out=False
6 )
7 X_train = ct.fit_transform(X_train_raw)

```

1 X_train_raw

	Exposure	Area	VehPower	VehBrand
0	0.01	A	4.0	0.0
...	...	...	...	...
508508	0.42	D	6.0	1.0

508509 rows × 10 columns

1 X_train

	VehGas_Regular	VehBrand_B1
0	1.0	0.0
...	...	...
508508	0.0	0.0

508509 rows × 39 columns

Aside: The order of the variables matters

```
1 X_train_raw["Area_In_Words"] = X_train_raw["Area"].map({
2     "A": "Very low", "B": "Low", "C": "Medium",
3     "D": "High", "E": "Very high", "F": "Extremely high"
4 })
```

```
1 enc_wrong = OrdinalEncoder()
2 enc_wrong.fit(X_train_raw[["Area_In_Words"]])
3 enc_wrong.categories_
```

```
[array(['Extremely high', 'High', 'Low', 'Medium', 'Very high', 'Very low'],
      dtype=object)]
```

```
1 for i, area in enumerate(enc_wrong.categories_[0]):
2     print(f"The Area value {area} gets turned into {i}.")
```

The Area value Extremely high gets turned into 0.

The Area value High gets turned into 1.

The Area value Low gets turned into 2.

The Area value Medium gets turned into 3.

The Area value Very high gets turned into 4.

The Area value Very low gets turned into 5.

```
1 print(f"In other words, {' < '.join(enc_wrong.categories_[0])}.")
```

In other words, Extremely high < High < Low < Medium < Very high < Very low.

Aside: Manually set the order

```
1 ordered = ["Very low", "Low", "Medium", "High", "Very high", "Extremely high"]
2 enc_right = OrdinalEncoder(categories=ordered)
3 enc_right.fit(X_train_raw[["Area_In_Words"]])
4 enc_right.categories_
```

```
[array(['Very low', 'Low', 'Medium', 'High', 'Very high', 'Extremely high'],
      dtype=object)]
```

```
1 for i, area in enumerate(enc_right.categories_[0]):
2     print(f"The Area value {area} gets turned into {i}.")
```

The Area value Very low gets turned into 0.

The Area value Low gets turned into 1.

The Area value Medium gets turned into 2.

The Area value High gets turned into 3.

The Area value Very high gets turned into 4.

The Area value Extremely high gets turned into 5.

```
1 print(f"In other words, {' < '.join(enc_right.categories_[0])}.")
```

In other words, Very low < Low < Medium < High < Very high < Extremely high.

```
1 X_train_raw = X_train_raw.drop("Area_In_Words", axis=1)
```

Lecture Outline

- Standardising the numerical covariates
- Handling categorical variables
- **Missing values and rare observations**
- Procedures to keep your data organised

Stroke prediction dataset

Dataset source: [Kaggle Stroke Prediction Dataset](#).

```
1 RAW_DATA_DIR = Path("stroke/raw")
2 data = pd.read_csv(RAW_DATA_DIR / "stroke.csv")
3 data.head()
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12

Data description

1. **id**: unique identifier
2. **gender**: “Male”, “Female” or “Other”
3. **age**: age of the patient
4. **hypertension**: 0 or 1 if the patient has hypertension
5. **heart_disease**: 0 or 1 if the patient has any heart disease
6. **ever_married**: “No” or “Yes”
7. **work_type**: “children”, “Govt_job”, “Never_worked”, “Private” or “Self-employed”
8. **Residence_type**: “Rural” or “Urban”
9. **avg_glucose_level**: average glucose level in blood
10. **bmi**: body mass index
11. **smoking_status**: “formerly smoked”, “never smoked”, “smokes” or “Unknown”
12. **stroke**: 0 or 1 if the patient had a stroke

Look for missing values and split the data

First, look for missing values.

```
1 number_missing = data.isna().sum()
2 number_missing[number_missing > 0]
```

```
bmi      201
dtype: int64
```



Tip

Always take a look at the missing data in, say, Excel. Sometimes pandas reads in a category like “None” and interprets that as a missing value, when it’s really a valid category. This happened to me just the other day.

```
1 features = data.drop(["id", "stroke"], axis=1)
2 target = data["stroke"]
3
4 X_main_raw, X_test_raw, y_main, y_test = train_test_split(
5     features, target, test_size=0.2, random_state=7)
6 X_train_raw, X_val_raw, y_train, y_val = train_test_split(
7     X_main_raw, y_main, test_size=0.25, random_state=12)
8
9 X_train_raw.shape, X_val_raw.shape, X_test_raw.shape
```

```
((3066, 10), (1022, 10), (1022, 10))
```

What values do we see in the data?

```
1 X_train_raw["gender"].value_counts()
```

```
gender
Female    1802
Male      1264
Name: count, dtype: int64
```

```
1 X_train_raw["ever_married"].value_counts()
```

```
ever_married
Yes      2007
No       1059
Name: count, dtype: int64
```

```
1 X_train_raw["Residence_type"].value_count
```

```
Residence_type
Urban    1536
Rural    1530
Name: count, dtype: int64
```

```
1 X_train_raw["work_type"].value_counts()
```

```
work_type
Private      1754
Self-employed  490
children     419
Govt_job     390
Never_worked  13
Name: count, dtype: int64
```

```
1 X_train_raw["smoking_status"].value_count
```

```
smoking_status
never smoked  1130
Unknown      944
formerly smoked  522
smokes       470
Name: count, dtype: int64
```

Group sparse categories

Look for sparse categories. One common trick is to lump together all the small categories (e.g. < 5% or 10% each) into a new combined 'OTHER' or 'RARE' category. The [OneHotEncoder](#) can group infrequent levels for us.

```
1 X_train_raw["work_type"].value_counts()
```

```
work_type
Private      1754
Self-employed  490
children     419
Govt_job     390
Never_worked  13
Name: count, dtype: int64
```

```
min_frequency max_categories
```

Lump together any level below 5% of the rows:

```
1 enc = OneHotEncoder(sparse_output=False,
2   min_frequency=0.05)
3 ohe = enc.fit_transform(X_train_raw[["work_type"]])
4 enc.infrequent_categories_
```

```
[array(['Never_worked'], dtype=object)]
```

```
1 ohe.head(3)
```

	work_type_Govt_job	work_type_Private	work_type_Self-employed	work_ty
1914	0.0	1.0	0.0	0.0
1958	0.0	0.0	0.0	1.0
2920	0.0	1.0	0.0	0.0

Make a plan for preprocessing each column

1. Take categorical columns $\hookrightarrow$ one-hot vectors
2. binary columns $\hookrightarrow$ do nothing (already 0 & 1)
3. continuous columns $\hookrightarrow$ impute NaNs & standardise.

Make the column transformer

```

1  nom_vars = ["gender", "ever_married", "Residence_type",
2             "work_type", "smoking_status"]
3
4  ct = make_column_transformer(
5      (OneHotEncoder(sparse_output=False, handle_unknown="ignore"), nom_vars),
6      ("passthrough", ["hypertension", "heart_disease"]),
7      remainder=make_pipeline(SimpleImputer(), StandardScaler()),
8      verbose_feature_names_out=False
9  )
10
11 X_train = ct.fit_transform(X_train_raw)
12 X_val = ct.transform(X_val_raw)
13 X_test = ct.transform(X_test_raw)
14
15 for name, X in zip(("train", "val", "test"), (X_train, X_val, X_test)):
16     num_na = pd.DataFrame(X).isna().sum().sum()
17     print(f"The {name} set has shape {X.shape} & with {num_na} NAs.")

```

The train set has shape (3066, 20) & with 0 NAs.

The val set has shape (1022, 20) & with 0 NAs.

The test set has shape (1022, 20) & with 0 NAs.

Handling unseen categories II

This was caused by fitting the encoder on the `X_train` which, by bad luck, didn't have the "Other" category. Why not just read the entire dataset before splitting to collect all the values the categorical variables could take?

It's a very mild form of data leakage, though it may be justifiable in some cases. If we made the datasets, we'd have a better idea of whether it makes sense on a case-by-case basis. We struggle here as we (in this course) are just analysing datasets which we didn't collect.

Lecture Outline

- Standardising the numerical covariates
- Handling categorical variables
- Missing values and rare observations
- **Procedures to keep your data organised**

Directory structure

A popular convention (from [Cookiecutter Data Science](#)) splits `data/` by *how far along* the data is:

```
data/
├── raw/          # original, never edited
├── interim/      # cleaned, still readable
└── processed/    # final, model-ready
```

- **raw/** is *immutable* — never overwrite it, so the whole pipeline can always be rebuilt from scratch.
- **interim/** is data that can be recreated from raw & a preprocessing script.
- **processed/** is what the model loads: split, encoded and scaled.

interim/ holds data that is **nearly model-ready but still human-readable** — after cleaning, joining and splitting, but *before* rescaling strips the units and one-hot encoding turns everything into a wall of 0s and 1s.

- You can **open it and eyeball it**: ages still look like ages, premiums still in dollars.
- It **caches** slow steps (parsing, joining, geocoding) so tweaking the final features doesn't re-run them.
- It keeps **processed/** **unambiguous** — one obvious file per train/validation/test split.

Save the preprocessed data to files

```
1 PROCESSED_DATA_DIR = Path("stroke/processed")
2 PROCESSED_DATA_DIR.mkdir(parents=True, exist_ok=True)
3
4 X_train.to_csv(PROCESSED_DATA_DIR / "x_train.csv", index=False)
5 X_val.to_csv(PROCESSED_DATA_DIR / "x_val.csv", index=False)
6 X_test.to_csv(PROCESSED_DATA_DIR / "x_test.csv", index=False)
7 y_train.to_csv(PROCESSED_DATA_DIR / "y_train.csv", index=False)
8 y_val.to_csv(PROCESSED_DATA_DIR / "y_val.csv", index=False)
9 y_test.to_csv(PROCESSED_DATA_DIR / "y_test.csv", index=False)
```

Keep the preprocessor around

Later want to make a prediction on some new data point. It has to go through the **exact same** preprocessing steps.

```
1 joblib.dump(ct, PROCESSED_DATA_DIR / "preprocessor.pkl")
2 preprocessor = joblib.load(PROCESSED_DATA_DIR / "preprocessor.pkl")
```

```
1 X_test_raw.head(1)
```

	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bp
2804	Female	69.0	0	0	Yes	Govt_job	Rural	70.98	3C

```
1 new_data = X_test_raw.head(1).copy()
2 new_data["ever_married"] = "No" # Let's pretend this is the new data point
```

```
1 ct.transform(new_data)
```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_Urban
2804	1.0	0.0	1.0	0.0	1.0	0.0

```
1 preprocessor.transform(new_data)
```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_Urban
2804	1.0	0.0	1.0	0.0	1.0	0.0

Glossary

- column transformer
- nominal variables
- ordinal variables
- one-hot encoding
- missing values
- imputation
- raw / interim / processed data
- standardisation